Fish have streamlined body structure which helps them to move smoothly with the flow of water. Muscles and fins on the body and the tail help to keep the balance.

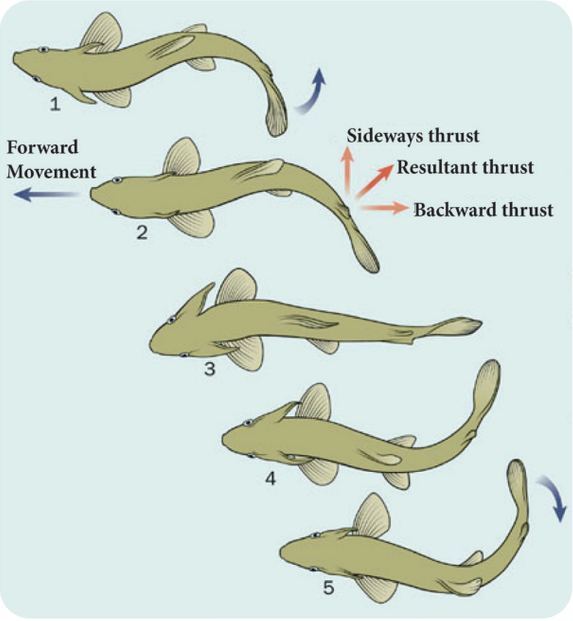


Figure 19.5 Movement in Fish (Swimming)

Activity 4

Make a paper boat; put it in water and push it with narrow end pointing forward. Now hold the boat sideway and push it into water from the broad side. What did you observe? In which process was it easy to move the boat? Have you noticed that the shape of a boat is somewhat like a fish?

19.2.6 Movements in Human body

Humans can move some parts of their body in different directions; however some body parts can be moved only in one direction. Our body is made up of a frame work of bones called **skeleton** which helps in the movement of the body. Some of the movements in body parts of human are:

a. Movement of eyelids
b. Movement of the heart muscles
c. Movement of jaw
d. Movement of arms and legs
e. Movements of head
f. Movements of neck

Movement of some organs happens because of the combined action of bones and muscles. In such cases, movement is possible along a point where two or more bones meet.

- Cheetah can run 76 kilometrer per hour.
- Hippopotamus can run faster than a man.
- Cockroach is the fastest animal with 6 legs covering a metre per second.
- The fastest mammal, the dolphin can swim upto 35 miles per hour.

19.3 Types of Movements

When we talk about locomotion and movement, there are three types of movements.

19.3.1 Amoeboid movement

It is brought about by pseudopodia which are appendages which move with the movement of protoplasm within a cell.

19.3.2 Ciliary movement

This movement is brought about by appendages called as cilia which are the hair-like extensions of the epithelium. Both these kinds of movements are seen with the cells of the lymphatic system.

19.3.3 Muscular movement

It is a more complex movement which is brought about by the musculoskeletal system. This type of movement is seen in the higher vertebrates.

To understand more about the movements brought about by the musculoskeletal system, we need to understand the joints, skeleton and types of muscles.

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